Jared Marx-Kuo

Eckhart 304, University of Chicago, Chicago, IL
jmarxkuo@gmail.com https://jmarxkuo.github.io/

EMPLOYMENT

NSF Postdoc and Dickson Instructor, University of Chicago Math Department — 2026 – Present

Postdoctoral Visitor, Center for Mathematical Modeling, Santiago, Chile — June 2026

NSF Postdoc, Rice University Math Department — 2025 – 2026

Griffith Conrad Evans Instructor, Rice University Math Department — 2024 – 2025

EDUCATION

PhD in Mathematics, Stanford University — 2019 – 2024

AWARDS

NSF Mathematical Sciences Postdoctoral Research Fellowship, National Science Foundation — 2025 – 2028

Rice Academy of Fellows, Rice University — 2025 – 2026
- Received $\$2,000$ professional development grant

Rice Creative Ventures Fund Grant, Rice University — October 2025
- Received $\$10,000$ to host a conference on geometry and dynamics in April 2026
- Joint with David Fisher, Francisco Arana-Herrara, Christos Mantoulidis

US Junior Oberwolfach Fellow 2025, MFO — July 2025
- Received $\$1000$ to attend Oberwolfach workshop Differentialgeometrie im Großen 2025

Stanford CTL LIT Fellow, Stanford Center for Teaching and Learning — 2023
- $2000 grant to develop TA handbook + workshops addressing: active learning, instructor talk, and identity in the classroom
- Joint with Alexandra Stavrianidi

NSF Graduate Research Fellow, National Science Foundation — 2021
- Full Funding for 3 years of PhD

Stanford Graduate Fellowship in Science & Engineering, Stanford University — 2019
- Full Funding for 3 years of PhD

PUBLICATIONS

The p-widths of the Hemisphere
(Accepted at **Communications in Analysis and Geometry**)

Almgren's Three-Legged Starfish (Joint with Christos Mantoulidis)
(2025, To appear at the **Proceedings of the AMS**)

"The p-widths of $\mathbb{RP}^2$"
Proceedings of the AMS, Vol 153, No. 12 (2025) 5385 – 5396

"Diameter, Area, and Mean Curvature" (Joint with Gregory Chambers)
Differential Geometry and its Applications, Vol 103 (2026)

"Index, Intersections, and Multiplicity of Min-Max Geodesics" (Joint with Lorenzo Sarnataro, Douglas Stryker)
Advances in Mathematics, Volume 502, Part C. Article 111193

"The Isospectral Problem for p-widths: An Application of Zoll metrics"
Comptes Rendus Mathématique, Vol 363 (2025), 565 – 570

"An Inverse Problem for Renormalized Area/Entanglement Entropy"
New Zealand Journal of Mathematics, Vol 56 (2025) 69 – 124
(Video on this paper from BIRS 2024)

"Second Inner Variations, Stress-Energy Tensors, and Index of Limiting Varifolds"
Journal of Geometric Analysis Vol 35, No. 290 (2025)

"Geometric Variations of an Allen–Cahn Energy on Hypersurfaces" (Joint with Èrico Melo Silva)
Calculus of Variations and Partial Differential Equations, Vol 64, No. 180 (2025)

"A Dirichlet-to-Neumann Map for the Allen-Cahn Equation on Manifolds with Boundary"
Interfaces and Free Boundaries, Vol 27, No. 4 (2025) 575 – 618

"Variations of Renormalized Volume for Minimal Submanifolds of Poincare-Einstein Manifolds"
Communications in Analysis and Geometry, Vol 33, No. 1 (2025) 17 – 29

"Sandpile Groups of Cayley Graphs of $\mathbb{F}_2^N$" (Joint with Jiyang Gao, Vaughan McDonald, Chi Ho Yuen)
Communications in Algebra, Vol 52, No. 10 (2024), 4459 – 4479

"Characters of Renner Monoids and Their Hecke Algebras" (Joint with Andy Hardt, Vaughan McDonald, John M. O'Brien, Alexander Vetter)
International Journal of Algebra and Computation, Vol 30, No. 7 (2020), 1505 – 1535

PREPRINTS

Equivariant Allen–Cahn Solutions and the Existence of Cohomogeneity 2 Minimal Surfaces (Joint with Rayssa Caju and Pedro Gaspar)
(2026, Submitted)

Mountain Pass Critical Points of the Volume Constrained Area Functional (Joint with Gregory R. Chambers)
(In Revision)

"Infinitely Many Surfaces with Prescribed Mean Curvature in the Presence of a Strictly Stable Minimal Surface" (Joint with Pedro Gaspar)
(2025, Submitted)

"Convolution Algebras for Finite Reductive Monoids" (Joint with Vaughan McDonald, John M. O'Brien, Alexander Vetter)

TEACHING

Math 355, Instructor, *Linear Algebra*, Rice — January 2025 – April 2025

Math 354, Instructor, *Honors Linear Algebra*, Rice — September 2024 – December 2024

Math 53 TA, *Ordinary Differential Equations with Linear Algebra*, Stanford — March 2023 – June 2023

LSP Math Instructor, Leland Scholars program, Stanford — August 2021
- Instructor for course on high school level mathematics for incoming FGLI freshmen
- Gave welcome talk, "*Mathematics and Society*"

Math 62CM TA, *Modern Mathematics: Continuous Methods*, Stanford — January 2021 – March 2021

Math 171 CA, *Fundamental Concepts of Analysis*, Stanford — September 2020 – November 2020

SUMAC TA, Stanford University Mathematics Camp, Stanford — June 2020 – August 2020
- Teaching Assistant for summer course in algebraic topology
- Gave talk, "*Math, Science, and Society*"

TALKS

Recent Advances on Geometric Variational Problems and Curvature, JMM Special Session — *January 2027*
Geometric Inequalities, Convergence, and Scalar Curvature, JMM Special Session — *January 2027*
Analysis and Mathematical Physics Seminar, Institute of Advanced Study — November 2026
UMD Geometry Seminar, University of Maryland College Park — *October 2026*
PUC Geometry Seminar, Pontificia Universidad Catolica de Chile — *June 2026*
IMPA Geometry Seminar, Instituto Nacional de Matematica Pura e Aplicada — *June 2026*
CMM Analysis and PDE Seminar, Center for Mathematical Modeling (Santiago, Chile) — *June 2026*
UTD Colloquium, University of Texas Dallas — *March 2026*
Toronto Geometry Seminar, University of Toronto — *March 2026*
Cal Tech Geometry Seminar, California Institute of Technology — *February 2026*
UC Irvine Geometry Seminar, University of California Irvine — *February 2026*
Yale Geometry Seminar, Yale University — *February 2026*
BIP Inverse Problems Online Seminar, (Hosted by Sebastian Munoz-Thon) — *October 2025*
UPenn Geometry Seminar, University of Pennsylvania — *October 2025*
Lehigh University Geometry Seminar, Lehigh University — *October 2025*
Rice University Undergraduate Colloquium, Rice University — *October 2025*

Johns Hopkins University Geometry Seminar, Johns Hopkins University *September 2025*
Rutgers University Geometry Seminar, Rutgers University *September 2025*
Cornell University Geometry Seminar, Cornell University *September 2025*
Oberwolfach Differentialgeometrie im Großen, MFO *July 2025*
UC Santa Barbara Geometry Seminar, University of Santa Barbara California *May 2025*
UC Santa Cruz Geometry and Analysis Seminar, University of Santa Cruz California *May 2025*
Weyl Laws Across Mathematics, Brin Mathematics Research Center *May 2025*
UTK Geometry Seminar, University of Tennessee Knoxville *March 2025*
Conformal and CR Geometry Workshop, BIRS (Video) *December 2024*
Rice Geometry Seminar, Rice University *September 2024*
Cornell Geometry Seminar, Cornell University *April 2024*
Syracuse Geometry Seminar, Syracuse University *April 2024*
UChicago Geometry Seminar, University of Chicago *April 2024*
Northwestern University Analysis Seminar, Northwestern University *March 2024*
Calculus of Variations and Geometric Measure Theory Seminar, University of Pisa (Zoom) *March 2024*
UC Davis Geometry Seminar, University of California Davis *December 2023*
Rice Geometry Seminar, Rice University *September 2023*
Early Career Seminar on Inverse Problems, PDEs, Microlocal Analysis, Differential Geometry, University of Washington *March 2023*
UC Santa Cruz Geometry Seminar, University of California Santa Cruz *November 2022*
UC Davis Geometry Seminar, University of California Davis *October 2022*
UMD Geometry Seminar, University of Maryland College Park *September 2022*
Stanford Geometry Seminar, Stanford University *September 2022*
Conformal Geometry, Analysis, and Physics Conference in Honor of Robin Graham, University of Washington (Young researcher poster session) *June 2022*

Outreach and service

Referee service (Periodic)

- Inventiones Mathematicae
- Journal of Differential Geometry
- Transactions of the American Mathematical Society
- Selecta Mathematica
- Nonlinear Analysis
- Calculus of Variations and PDEs
- Mathematische Zeitschrift

University of Chicago Geometry Seminar Organizer September 2026 - Present

University of Chicago Directed Reading Program Mentor September 2026 - Present

Rice Summer Undergraduate Research Fellowship (SURF) Mentor Program Website June - July 2025

- Individual reading with a student in Complex Analysis for 2 months

Oberwolfach Differentialgeometrie im Großen 2025 Reporter MFO July 2025

- Workshop reporter for “Differentialgeometrie im Großen" 2025

Rice University Geometry Seminar Organizer, September 2024 - May 2026

Rice University Math Ed Discussion Seminar, December 2024, September 2025

- Presented on Identity and Privilege in the math classroom
- Led an interactive workshop discussing scenarios + actionable items using Universal Design for Learning (UDL)
- Led a discussion on AI, homework, and teaching in math classes

Rice University Directed Reading Program Mentor, Program Website September 2024 - June 2025

Stanford TA Workshop, Program Website September 2023 - June 2024

- Organized workshops on Active Learning, Instructor Talk, and Identity in the Classroom, to help TAs improve their teaching and student interactions
- Sponsored by the Stanford CTL LIT Fellowship (see Awards)

Stanford Directed Reading Program Mentor, Program Website September 2019 - June 2024

Stanford Directed Reading Program Organizer, Program Website January 2020 - June 2024

- Organized program and paired undergraduates with math graduate students for quarter-long, 1-on-1 reading projects
- Advertise the program, schedule presentations, submit budget updates, etc.

President of Stanford Graduate Math Outreach Organization (SGMOO), May 2022 - June 2024
- Organizer for outreach programs run by math graduate students, including: Directed Reading Program (DRP), Stanford Women in Math Mentoring Program (SWIMM)

Stanford First-Year Mentoring Program Mentor + Organizer, September 2020 - June 2023
- Paired older graduate students in math department with first year PhD students to facilitate bonding, research connections, and to provide PhD advice
- Participated as mentee in 2019, as mentor in 2020-2023

TA Mentor for First-time TAs 2022 - 2023
- Mentored new TAs - observed their section teaching and provided feedback
- Compensated role

President of Stanford Math Graduate Students September 2020 - September 2022
- Organized graduate student roles to develop a sense of community within the department

SUMAC Application Reader, Program Website March 2021 - April 2021
- Read applications for Stanford University Mathematics Camp
- Advocated for underrepresented and underresourced students in the application process

SURIM Mentor, Stanford Undergraduate Research Institute in Mathematics June 2020 – August 2020
- Mentored undergraduates in PDEs with applications to the Allen-Cahn Equation

SURA Mentor, Program Website January 2020 - April 2020
- Mentored undergraduates in math by advising them on courses and summer projects

FLI Community Mentor, Program Website January 2020 - November 2020
- Supported low income undergraduate student with navigating college life and adjusting plans for Covid-19 pandemic

REFERENCES

Andre Neves, Professor, Department of Mathematics, University of Chicago, aneves@math.uchicago.edu
Otis Chodosh, Associate Professor, Department of Mathematics, Stanford University, ochodosh@stanford.edu
Christos Mantoulidis, Associate Professor, Department of Mathematics, Rice University, christos.mantoulidis@rice.edu